Hall Ticket Number:

MR-21

09/02/24 AN

Code: B414CS3S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, Jan/Feb-2024

Database Management Systems

(Common to CSE & IT)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

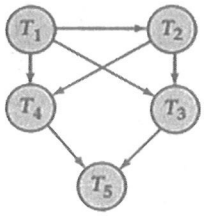
Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	EmployeeDetails EmpName | Branch | Address | Salary Raja | Kolkata | Kolkata | 10000 Priyanka | Mumbai | Hyderabad | 12000 Raja | Bangalore | Kolkata | 10000 Rumki | Hyderabad | Hyderabad | 20000 Sumit | Hyderabad | Bangalore | 10000 Sandip | Mumbai | Mumbai | 15000 Identify Candidate Key from the above relation.	2	2	1	1,2
1. b)	Write Database system Applications.	2	2	1	1,2
1. c)	What is Referential Integrity?	2	2	2	1,2
1. d)	Construct SQL queries that are equivalent to each of the following i) $\Pi_A(R)$ ii) R x S Where relational schemas are R=(A,B,C) S=(D,E,F)	2	3	2	1,2
1. e)	Define and explain First Normal Form with an example.	2	2	3	1,2
1. f)	List the problems caused by Redundancy.	2	1	3	1,2
1. g)	What are the benefits of a Two-Phase Locking Protocol?	2	2	4	1,2
1. h)	Write Transaction properties?	2	2	4	1,2
1. i)	Construct the B+ tree for the following set of key values with order 3. 5 15 25 35 45	2	3	5	1,2
1. j)	What is Dense Index? Give an Example.	2	1	5	1,2

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Explain the Three-Schema Architecture with diagram.	5	2	1	1,2
2. b)	Design an E-R Diagram for Airline Reservation System.	5	3	1	1,2,3
	OR				
2. c)	Design an E-R diagram for keeping track of the exploits of your favourite sports team. You should store the matches played, the scores in each match, the players in each match and individual player statistics for each match. Summary statistics should be modelled as derived attributes.	6	3	1	1,2,3
2. d)	Explain the Advantages of Database Management system over Traditional File Processing System.	4	2	1	1,2
3. a)	What are the Fundamental Relational Algebra Operations? Give an example for each operation.	4	2	2	1,2
3. b)	Write SQL Queries for the given Schemas Employee: (Emp_id, Ename, office, Age) Books: (isbn, title, author, publisher) Loan : (Emp_id, isbn, date) (i) Print the names of employees who have borrowed any book published by Pearson publisher. (ii) Print the names of employees who have borrowed all books published by Pearson publisher.	6	3	2	1,2,3

P.T.O

	(iii) For each publisher, print the names of employees who have borrowed more than five books of that publisher.				
	OR				
3. c)	What is Primary Key, Super Key and Candidate Key explain with an example.	4	2	2	1,2
3. d)	Consider the following Relational Database and write SQL queries: employee (person_name, street, city) works (person_name, company name, salary) company (company_name, city) a. Find the names of all employees who live in city "Miami". b. Find the names of all employees whose salary is greater than $10,000. c. Find the names of all employees who live in "Miami" and whose salary is greater than $500,000.	6	3	2	1,2,3
4. a)	Define Normalization and write the importance of Normalization.	5	2	3	1,2
4. b)	Suppose that data base schema R = (A, B, C, D, E, F) of explain and Apply 3NF AB → C A → D B → EF	5	2	3	1,2,3
	OR				
4. c)	Explain different types of joins with examples.	5	2	3	1,2
4. d)	Write rules to find closure set of functional dependencies. Given relation schema r (A, B, C, G, H, I) and the set of functional dependencies: A→B A→C CG→H CG→I B→H Compute at least one super for the above schema (Hint: use attribute closure).	5	3	3	1,2,3
5. a)	Compare Conflict Serializability and View Serializability with an example.	5	2	4	1,2
5. b)	Describe the anomalies in concurrent execution of Transactions.	5	2	4	1,2
	OR				
5. c)	Explain Multiple Granularity protocol.	5	2	4	1,2
5. d)	Explain Conflict Serializable schedule. Consider the precedence graph of Figure verify the corresponding schedule Conflict Serializable or not. 	5	3	4	1,2
6. a)	Write short notes on Indexing Techniques.	5	2	5	1,2
6. b)	Construct a B+ tree with fan-out (no of pointers per node) is 3 for the following search key values 80, 50, 30, 70, 30, 100, 90. Assume that the tree is initially empty, and the values are added in the order given. a) Show the tree after insertion of 10, after insertion of 30, and after insertion of 90. b) Show the tree after deletion of 30, 10.	5	3	5	1,2,3
	OR				
6. c)	Explain File Organization and Indexing.	5	2	5	1,2
6. d)	Show the Extendable Hash structure for the file which contains records with the following search key values: 2 3 5 7 12 17 19 23 29 31 Where Hash Function is h(x)=x mod 8 and bucket size is 3.	5	3	5	1,2,3

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome